

**Chapter 2a – Rational Expressions – REVIEW**  
MCR3U

**Section 1: Negative and Rational Exponents**

**1)** Evaluate. Express as a fraction in lowest terms.

**a)**  $10^{-1}$

**b)**  $4^{-2}$

**c)**  $3^{-2} + 9^{-1}$

**d)**  $5^{-3} + 5^0$

**e)**  $\left(\frac{1}{5}\right)^{-1}$

**f)**  $\left(\frac{3}{4}\right)^{-3}$

**2)** Simplify. Express your answers using only positive exponents.

**a)**  $(x^{-2})(x^{-1})(x^0)$

**b)**  $(3km^2)(2k^{-2}m^{-2})$

**c)**  $w^{-3} \div w^{-2}$

**d)**  $\frac{u^{-2}v^3}{u^{-3}v^{-2}}$

**e)**  $(z^{-3})^{-2}$

**f)**  $(2ab^{-1})^{-2}$

**3)** Simplify. Express your answers using only positive exponents.

**a)**  $(4a^{-2})(-2a^{-3})$

**b)**  $\frac{(2x^2y)^{-2}(3xy)^{-1}}{(6x^2y^2)^{-2}}$

**c)**  $\left(\frac{1}{4x^2}\right)^{-2}$

**d)**  $\left(\frac{6a^3}{4b^4}\right)^{-2}$

**4)** Evaluate.

**a)**  $\sqrt[3]{64}$

**b)**  $\sqrt[4]{625}$

**c)**  $\sqrt[5]{-3125}$

**d)**  $\left(\frac{1}{64}\right)^{\frac{1}{6}}$

**e)**  $27^{\frac{2}{3}}$

**f)**  $(-1000)^{\frac{4}{3}}$

**g)**  $-4^{-3}$

**h)**  $\left(\frac{3}{4}\right)^{-2}$

**i)**  $\left(-\frac{27}{125}\right)^{-\frac{2}{3}}$

**5)** Simplify. Express answers using only positive exponents.

**a)**  $n^{\frac{1}{2}} \times n^{\frac{1}{3}} \times n^{\frac{1}{4}}$

**b)**  $(27y^3)^{\frac{1}{3}} \times \left(\frac{1}{16y^4}\right)^{-\frac{3}{4}}$

**c)**  $(27x^6)^{\frac{2}{3}} \div (9x^4)^{\frac{1}{2}}$

**d)**  $\frac{x^{-\frac{2}{3}}}{x^{-\frac{4}{5}}}$

## **Section 2: Rational Expressions**

**6)** Simplify each expression and state all restrictions on  $x$ .

**a)**  $\frac{x+7}{x^2+10x+21}$

**b)**  $\frac{x^2-64}{x-8}$

**c)**  $\frac{x^2-9}{x^2-8x+15}$

**7)** Simplify each expression and state the restrictions.

**a)**  $\frac{3x^2}{5xy} \times \frac{20xy^3}{12xy}$

**b)**  $\frac{150a^3b^4}{20a^2b} \div \frac{6b}{8ab^2}$

$$\text{c)} \frac{1}{3x} + \frac{5}{2x^2}$$

$$\text{d)} \frac{4}{x-6} - \frac{3}{x-4}$$

**8)** Simplify each expression and state restrictions.

$$\text{a)} \frac{x^2+7x}{3x+21} \times \frac{x^2+3x+2}{x+2}$$

$$\text{b)} \frac{x^2+4x-60}{3x+30} \div \frac{x^2-8x+12}{6x-12}$$

$$\text{c)} \frac{3}{x^2+7x+10} - \frac{5x}{x^2-4}$$

$$\text{d)} \frac{-10x}{x^2+18x+32} + \frac{12x}{x^2+6x-160}$$

9) Simplify each expression and state any restrictions

a)  $\frac{x-8}{x+7} \times \frac{x+15}{x^2+12x-45}$

b)  $\frac{x^2+12x+20}{x+5} \div \frac{x^2+7x-30}{x+10}$

c)  $\frac{x+3}{x-7} - \frac{x+9}{x-2}$

d)  $\frac{x+8}{x+3} + \frac{x-6}{x^2+9x+18}$

e)  $\frac{5x+1}{2x-1} - \frac{3x-3}{1-2x}$